

Week 3 – Industry Investigation & Data Discovery

Project	Secure Retail – Product Catalogue System
Unit	Capstone B – Week 3
Investigation type	Option B – Online Investigation
Date	30 September 2026
Investigated by	Anup Koirala – Secure Retail Team
Comparison product	Apple iPhone 18 Pro Max (256GB)
Evidence	18 screenshots – see Appendix A

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Purpose and Method

In Capstone A our team built a **Frontend MVP** of the Secure Retail Product Catalogue. Before designing the database and backend for Capstone B, we investigated how real Australian retailers collect, display and use product data.

Method

1. Selected three online retailers that sell the same product categories as our project.
2. Searched each site for the **same product** (iPhone 18 Pro Max 256GB) so results could be compared fairly.
3. Reviewed the home page, search results, filters, product page, cart and wishlist, delivery, policies and help links.
4. Captured **18 screenshots** as evidence (Figures 1–18 in Appendix A).
5. Extracted the data items used and compared each system with our MVP.

Prices and stock were recorded on 30 September 2026 and may have changed since.

Task 1 – Organisations Investigated

Comparison at a glance

	JB Hi-Fi	Officeworks	Amazon AU
Type	Electronics retailer	Office & tech retailer	Online marketplace
Search results	16	59 (incl. accessories)	1,000+
256GB price	\$2,299	\$2,297	\$2,297
Price range	\$2,299 – \$4,699	\$2,297 – \$4,697	From \$2,297
Rating	2.7★ (7 reviews)	No reviews yet	2.4★ (3 ratings)
Filters	Price only	Availability, Category, Price	12+ filter groups
Stock / fulfilment	Stock by postcode or location	Pick up in-store or Deliver	Out-of-stock pre-order
Returns	Faulty goods and warranties guide	365 days (OnePass)	30 days change of mind
Standout feature	Price-match ("JB Deal")	AI assistant "Ask Ollie"	Rating breakdown + Verified Purchase

JB Hi-Fi (Figures 1–7)

- Every storage option shows **its own price**.
- **Stock availability** is checked by postcode or current location.
- Detailed product data: **model**, **SKU** and a **30+ field specifications table**.
- Trust features: price-match, live chat, phone support, scam alerts and 8 payment methods.
- Business rule: **"Limit 2 per customer"**.

Officeworks (Figures 8–13)

- Customers choose **Pick up in-store** or **Deliver to me** before adding to cart.
- Filters include **Availability**: Delivery, In Store, Click & Collect.
- Product information is split into **Specifications**, **Customer Reviews**, **Q & As** and **Delivery**.
- Trust signals: Price Beat Guarantee, 365-day returns, product recalls notice.
- **"Ask Ollie"** AI shopping assistant.

Amazon Australia (Figures 14–18)

- The most filters: Prime, **delivery day**, condition (New / Renewed / Used), seller and rating.
- Reviews show a **star breakdown (%)** and **Verified Purchase** labels.
- Out-of-stock items can still be ordered; payment is taken **only when the item ships**.
- **Estimated delivery dates** appear in search results.
- Sponsored results and "Best Seller" badges appear in search results.

Task 2 – Data Discovery

25 data items were identified. Items marked **(New)** were not in our Week 2 Data Inventory.

#	Data Item	Why Is It Important?	Who Uses It?	Evidence
1	Product Name / Title	Identifies the product in search results	Customers, Staff	Fig 1, 8, 14
2	Product Price	Supports purchasing decisions	Customers, Finance	Fig 2, 10, 16
3	Variant Price (New)	Shows the cost difference between options	Customers	Fig 2, 10, 16
4	Product Variants – colour, storage (New)	Customer picks the exact version	Customers, Inventory	Fig 2, 10, 16
5	SKU / Model / Product Code (New)	Uniquely identifies stock	Staff, Inventory system	Fig 2, 10
6	Product Images / Video	Customers see the product before buying	Customers	Fig 2, 10, 16
7	Specifications	Detailed feature comparison	Customers	Fig 5, 12, 16
8	Product Category	Browsing and filtering	Customers, Admin	Fig 8
9	Average Star Rating	Quick signal of quality	Customers, Managers	Fig 1, 10, 18
10	Review Count	Shows how reliable the rating is	Customers	Fig 1, 14
11	Rating Breakdown (New)	Shows the spread of opinions	Customers, Managers	Fig 18
12	Review Text + Verified Purchase (New)	Builds trust with real experiences	Customers, Admin	Fig 18
13	Stock Level / Availability	Prevents selling unavailable items	Customers, Inventory	Fig 4, 14, 17
14	Store Location / Postcode (New)	Local stock and delivery options	Customers, Logistics	Fig 4, 11
15	Fulfilment Method (New)	Delivery or Click & Collect	Customers, Store staff	Fig 8, 11
16	Delivery Cost (New)	Affects the total price	Customers, Finance	Fig 9, 14
17	Estimated Delivery Date (New)	Sets customer expectations	Customers, Logistics	Fig 14
18	Promotion / Discount	Drives sales	Customers, Marketing	Fig 1, 16
19	Purchase Limit per Customer (New)	Stops bulk buying of popular stock	System, Staff	Fig 3
20	Payment Methods	Flexible, trusted payment	Customers, Finance	Fig 2, 7, 13
21	Warranty Period (New)	Gives purchase confidence	Customers, Support	Fig 6
22	Returns Policy (New)	Reduces purchase risk	Customers, Support	Fig 9, 17
23	Seller / Shipper (New)	Shows who the customer buys from	Customers	Fig 17
24	Wishlist Items	Saves products and shows demand	Customers, Marketing	Fig 3, 10, 17
25	Order Number / Tracking Status	Lets customers follow their delivery	Customers, Support	Fig 7, 13

Task 3 – Module Investigation Summary

Module: Product Catalogue **Question:** What information helps customers choose products?

Group	What real systems show	Evidence
Price and value	Price for every variant, promotions, price-match promises, flexible payment (Afterpay, Zip, Latitude)	Fig 1, 2, 10, 11, 16
Product details	6–11 images plus video, key features, full specifications, SKU	Fig 2, 5, 12, 16
Social proof	Star rating, review count, rating breakdown, Verified Purchase, Q&A	Fig 1, 12, 18
Availability and delivery	Stock by postcode, pick up in-store, delivery cost and date	Fig 4, 11, 14
Finding and comparing	Search, filters, sort, compare, "frequently bought together"	Fig 1, 3, 8, 14

Key insight: The iPhone 18 Pro Max had **low ratings (2.4–2.7★)** on JB Hi-Fi and Amazon. Customers see this before buying, so review data has a strong effect on purchasing decisions.

Task 4 – User Insights

Note: The three people below are **fictional user personas** created from the Task 1–3 findings. They are **not real interview participants**. Real interviews will be added when completed.

Question	Maya, 21 – Student	Daniel, 44 – Small business owner	Margaret, 66 – Retiree
Profile	Shops on her phone, budget-conscious	Buys devices for 5 staff, values time	Less confident online, worried about scams
Information expected before purchase	Price of each option, reviews, Afterpay	Exact specs, SKU, warranty	Clear price, stock nearby
What builds trust	Many reviews, "verified purchase"	Returns policy, warranty, known seller	Phone number, secure payment logos
What saves time	Filters, sort by price	Stock and delivery date before checkout	Simple layout, Click & Collect
What is often missing	Honest reviews on new products	Multi-store stock, business invoices	Plain-English specifications
What would improve the experience	Price-drop alerts, saved wishlist	Order tracking, saved accounts	Live chat or phone help, larger text

Key themes

- All three need **clear price and stock information** before buying.
- **Trust signals differ by user type:** reviews (students), warranty and returns (business buyers), phone support and scam warnings (older users).
- **Time-saving features** matter to everyone: filters, delivery dates, Click & Collect.
- **Accessibility** (simple layout, readable text) is a gap in our MVP.

Task 5 – Data Gap Analysis

#	Our MVP	Industry System	Missing Opportunity
1	Featured products on the Home page only	Search with a results count	Product search
2	Browse by category only	Price, availability, brand, rating and condition filters	Product filters
3	No sorting	6–9 sort options	Sort options
4	Product cards only	Product page with gallery and specifications	Product detail page
5	One price per product	Colour and storage variants with their own price	Variants and variant pricing
6	No product code	Model, SKU and product code	SKU field
7	Store testimonials only	Product reviews with breakdown and Verified Purchase	Product reviews
8	No stock information	Stock by postcode, "Only 2 left in stock"	Stock level
9	No delivery information	Delivery cost, date and Click & Collect	Delivery options
10	Add to Cart (UI only)	Working cart plus Buy Now	Functional cart
11	Wishlist (UI only)	Wishlist saved to an account	Saved wishlist
12	No user accounts	Sign in, create account, password reset	Secure user accounts
13	No order tracking	"Track my order"	Order tracking
14	No compare feature	Compare / Add to Compare	Product comparison
15	Contact form only	Live chat, phone, AI assistant	More support channels
16	No policy information	Warranty, returns, secure transaction	Trust and policy information

Task 6 – Product Improvement Workshop

Improvements for **Capstone B – Version 2**, prioritised using **MoSCoW**:

Must Have	Should Have	Nice To Have
Product search	Filters and sort options	Product comparison
Product detail page with specs and SKU	Product reviews with rating breakdown	Stock by store and Click & Collect
Stock level on products	Delivery cost, estimate and returns info	"Frequently bought together"
Secure user accounts (hashed passwords, roles)	Wishlist saved to account	"Notify me when back in stock"
Functional cart stored in the database	Order tracking page	Live chat / AI assistant
Variants with a price per variant		"Buy Now" one-step purchase



Task 7 – Jira Items Created

Key	Type	Summary	Priority
SR-20	Research Finding	Week 3 industry investigation: JB Hi-Fi, Officeworks, Amazon AU	Medium
SR-21	Data Requirement	Add SKU, variants and variant price to product data	High
SR-22	Data Requirement	Add stock level / availability field to products	High
SR-23	Data Requirement	Add review data: rating, text, verified purchase, date	Medium
SR-24	User Story	As a customer, I want to search products so that I can quickly find what I need	High
SR-25	User Story	As a customer, I want a product page with specifications so that I can compare features	High
SR-26	User Story	As a customer, I want to see stock availability so that I don't order unavailable items	High
SR-27	User Story	As a customer, I want a secure account so that my cart and wishlist are saved	High
SR-28	Feature Request	Filters (price, category, availability) and sort options	Medium
SR-29	Feature Request	Product reviews and ratings with star breakdown	Medium
SR-30	Enhancement	Show delivery cost, estimate and returns policy on the product page	Medium
SR-31	Enhancement	Save wishlist to user account	Medium
SR-32	Feature Request	Product comparison and "frequently bought together"	Low
SR-33	Research Finding	User personas created; real user interviews still to be completed	Medium

Conclusion and Next Steps

Our Capstone A MVP shows **name, image, price and category**. Real retailers rely on much richer data: **variants, SKUs, specifications, reviews, stock and delivery information**. This investigation gives us evidence-based requirements for the Capstone B database and backend.

Next steps

- ☐ Complete real user interviews and add them to the report
- ☐ Update the Week 2 Data Inventory with the **(New)** data items
- ☐ Begin database design for Products, Variants, Reviews, Stock, Users and Orders
- ☐ Refine Jira items SR-21 to SR-27 into sprint-ready tasks

Appendix A – Evidence Screenshots

All screenshots were captured on 30 September 2026. Personal details (delivery location and reviewer names) have been blurred.

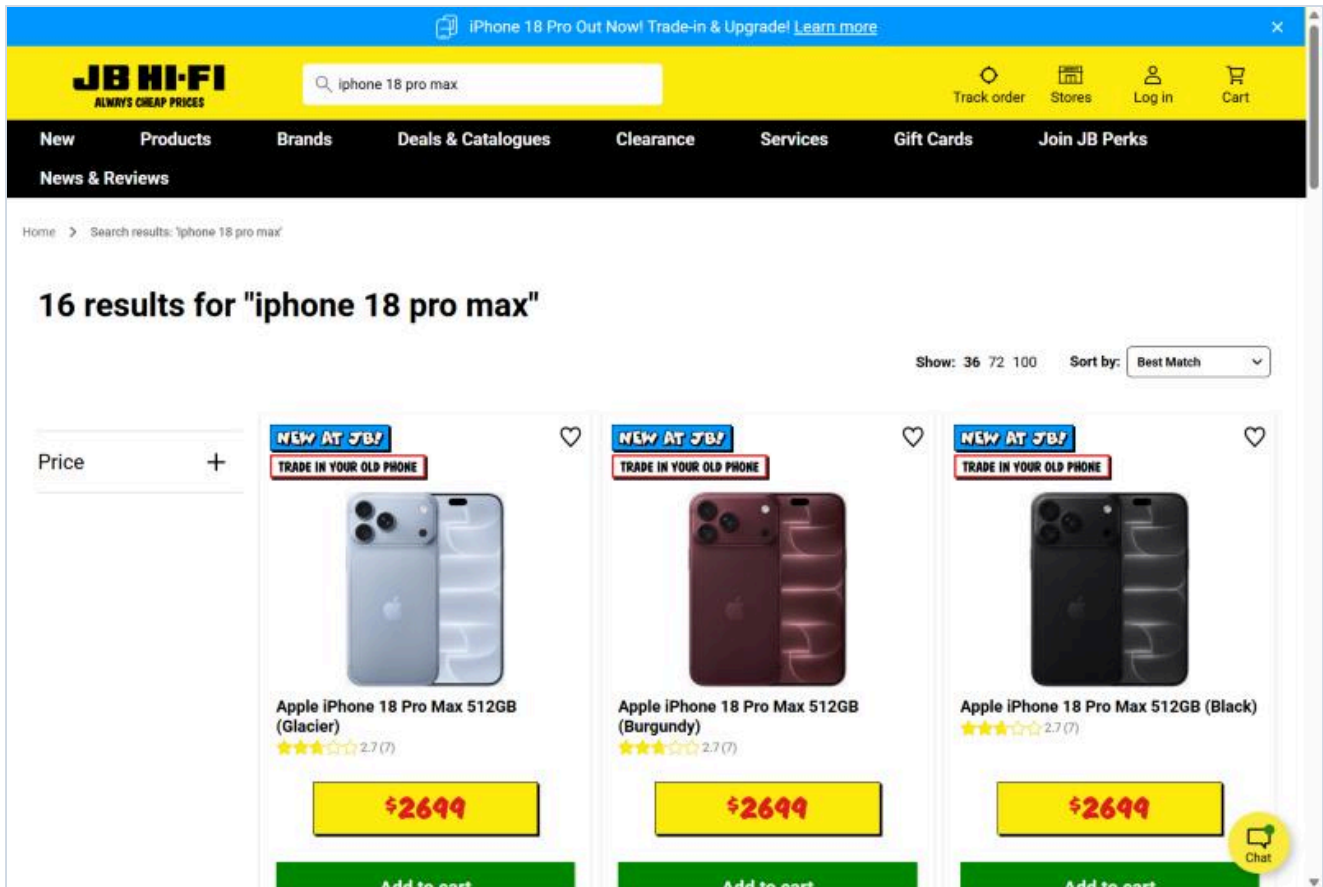


Figure 1 – JB Hi-Fi search results: menu, search bar, Track order/Stores/Log in/Cart, Price filter, sort and product cards

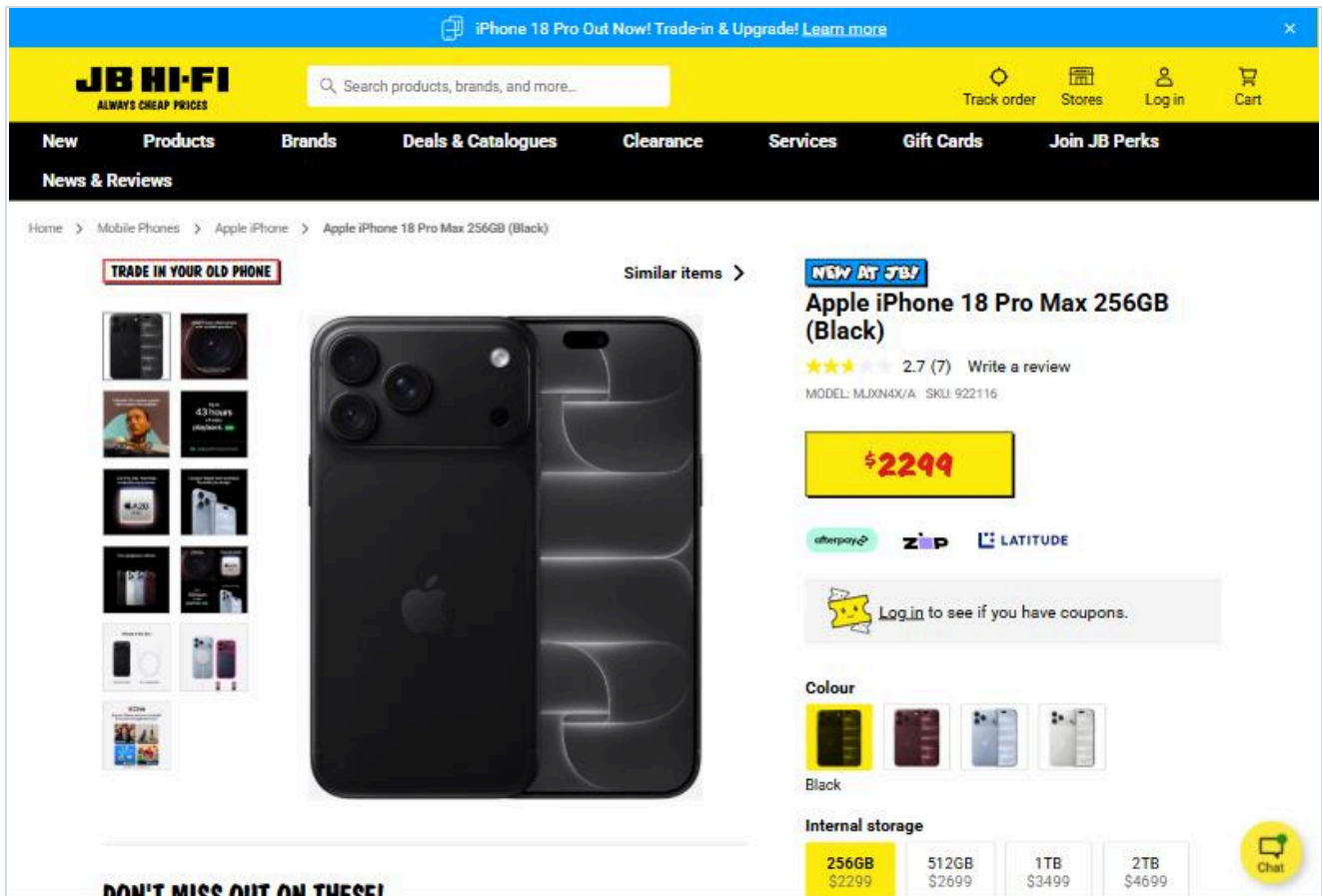


Figure 2 – JB Hi-Fi product page: price, rating, model/SKU, payment options, colour and storage options

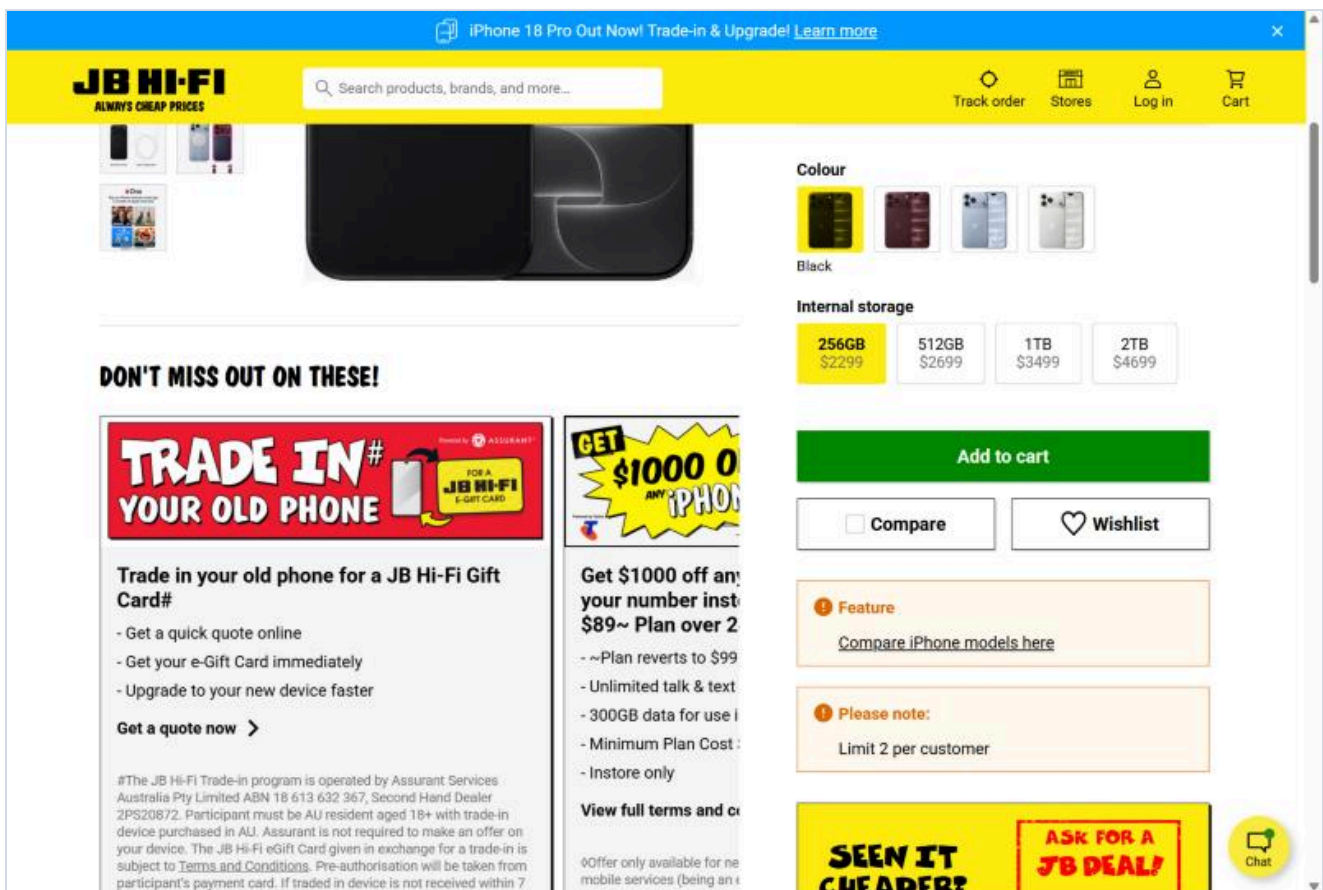


Figure 3 – JB Hi-Fi: Add to cart, Compare, Wishlist, 'Limit 2 per customer' and trade-in offer

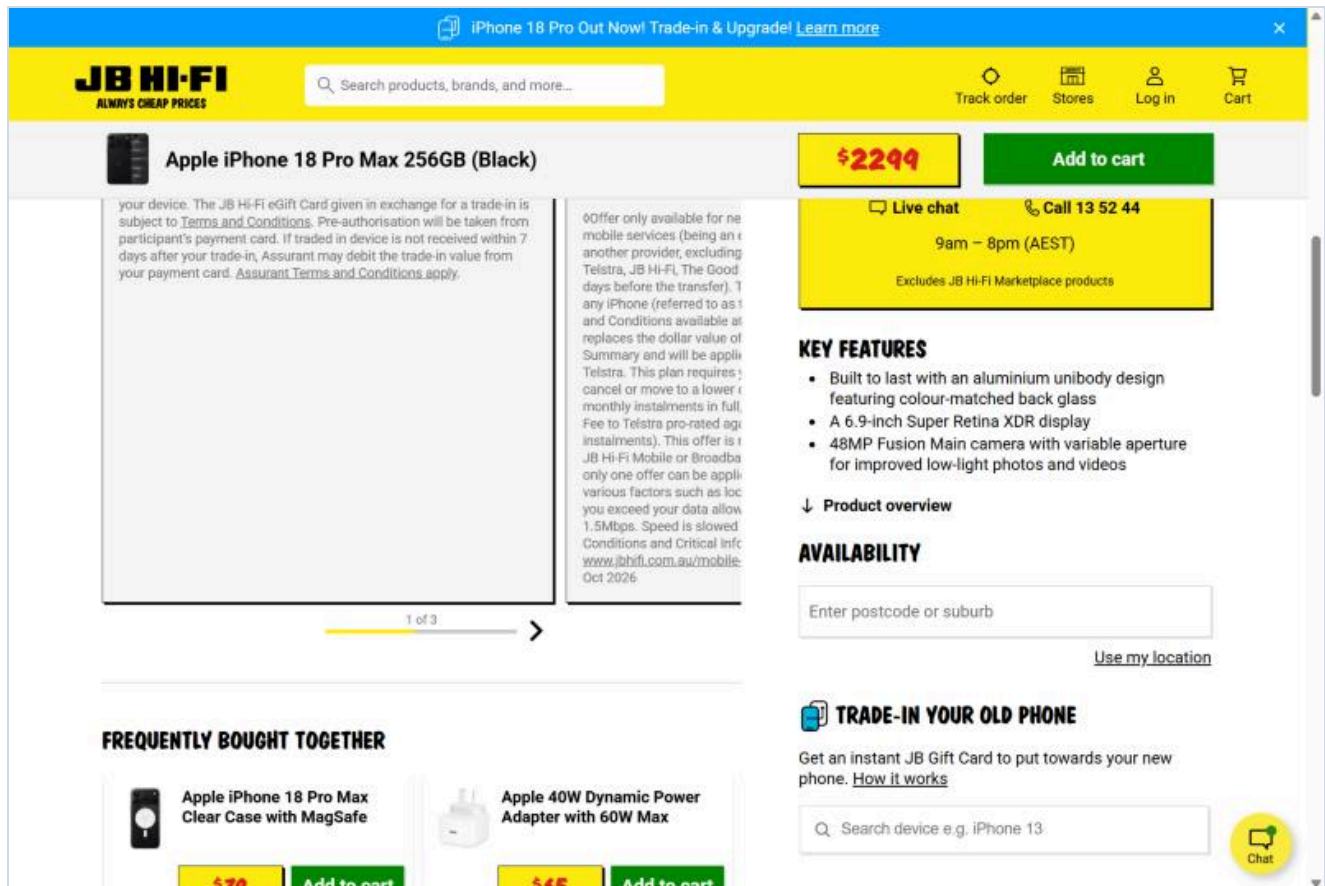


Figure 4 – JB Hi-Fi: live chat/phone, key features, stock availability by postcode, 'Frequently bought together'

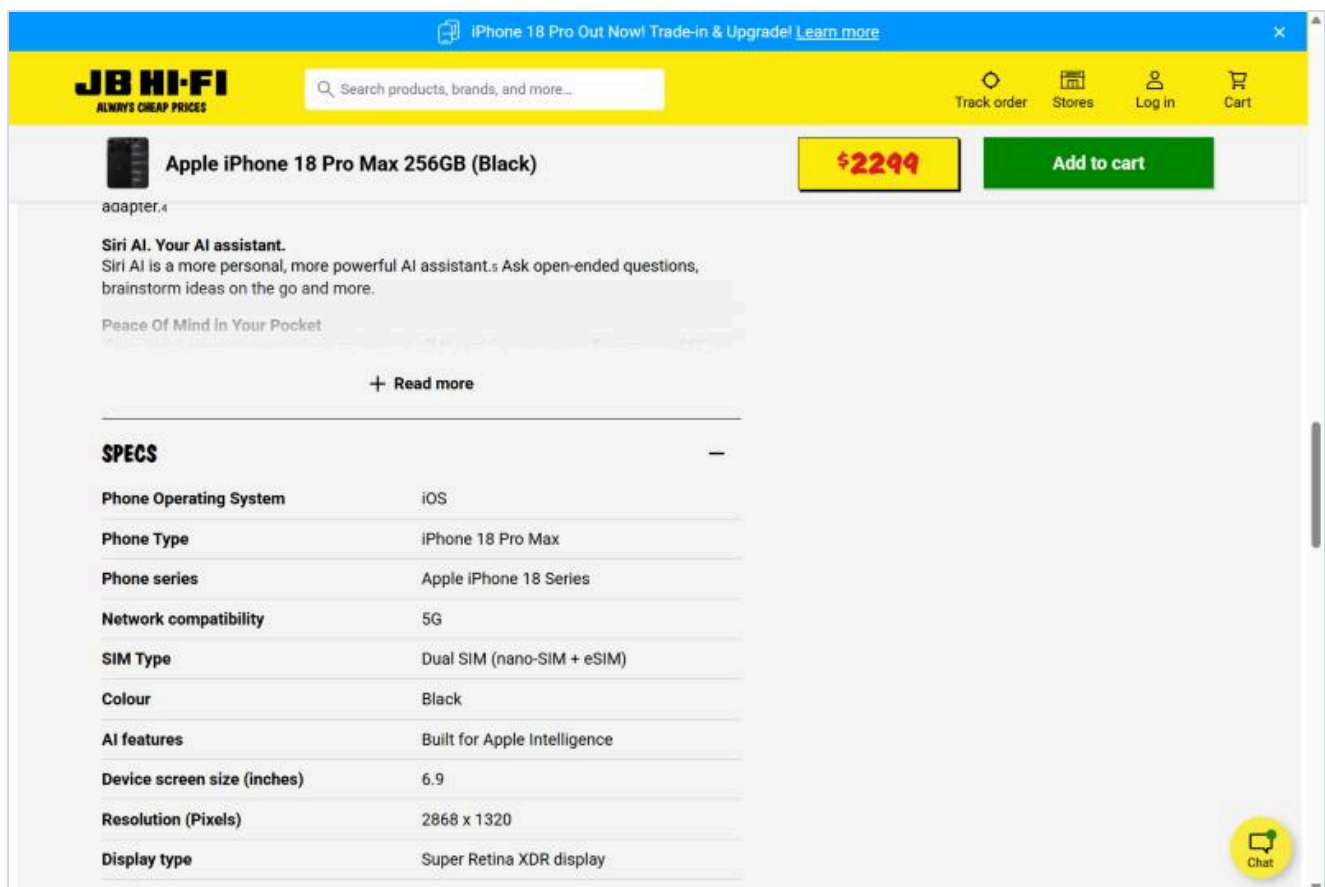


Figure 5 – JB Hi-Fi: specifications table

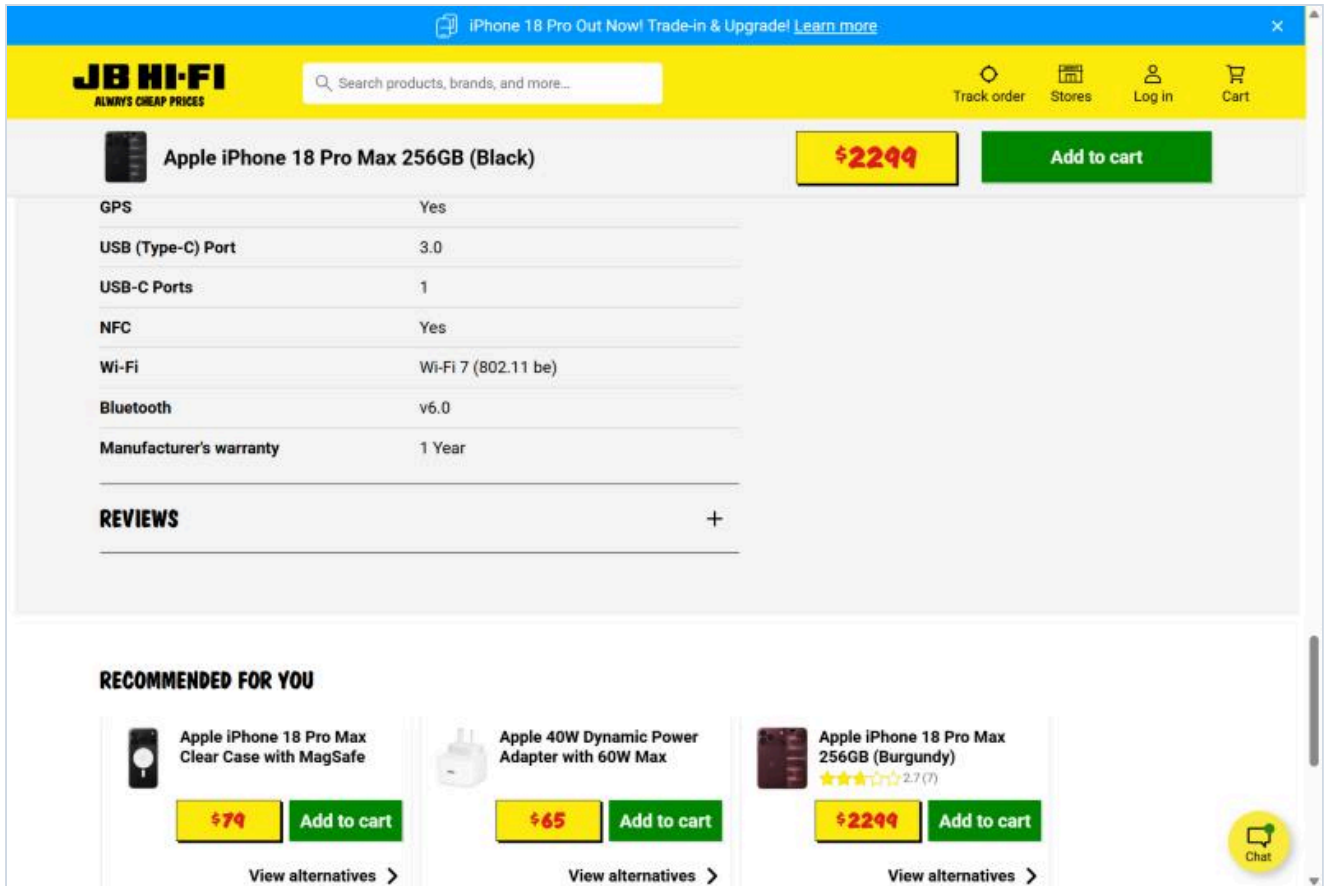


Figure 6 – JB Hi-Fi: 1-year manufacturer's warranty, Reviews section and 'Recommended for you'

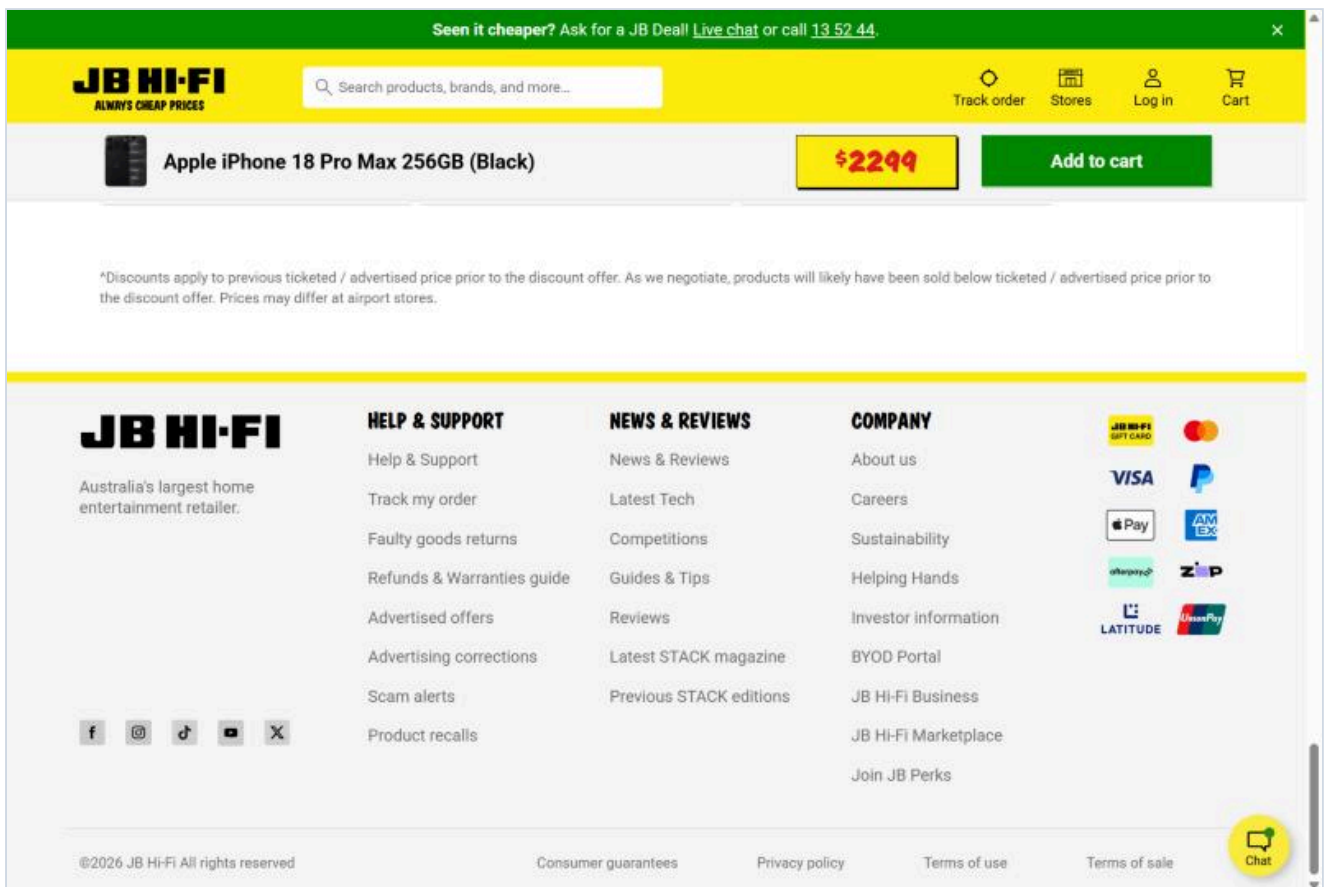


Figure 7 – JB Hi-Fi footer: Track my order, returns, warranties guide, scam alerts, accepted payment methods

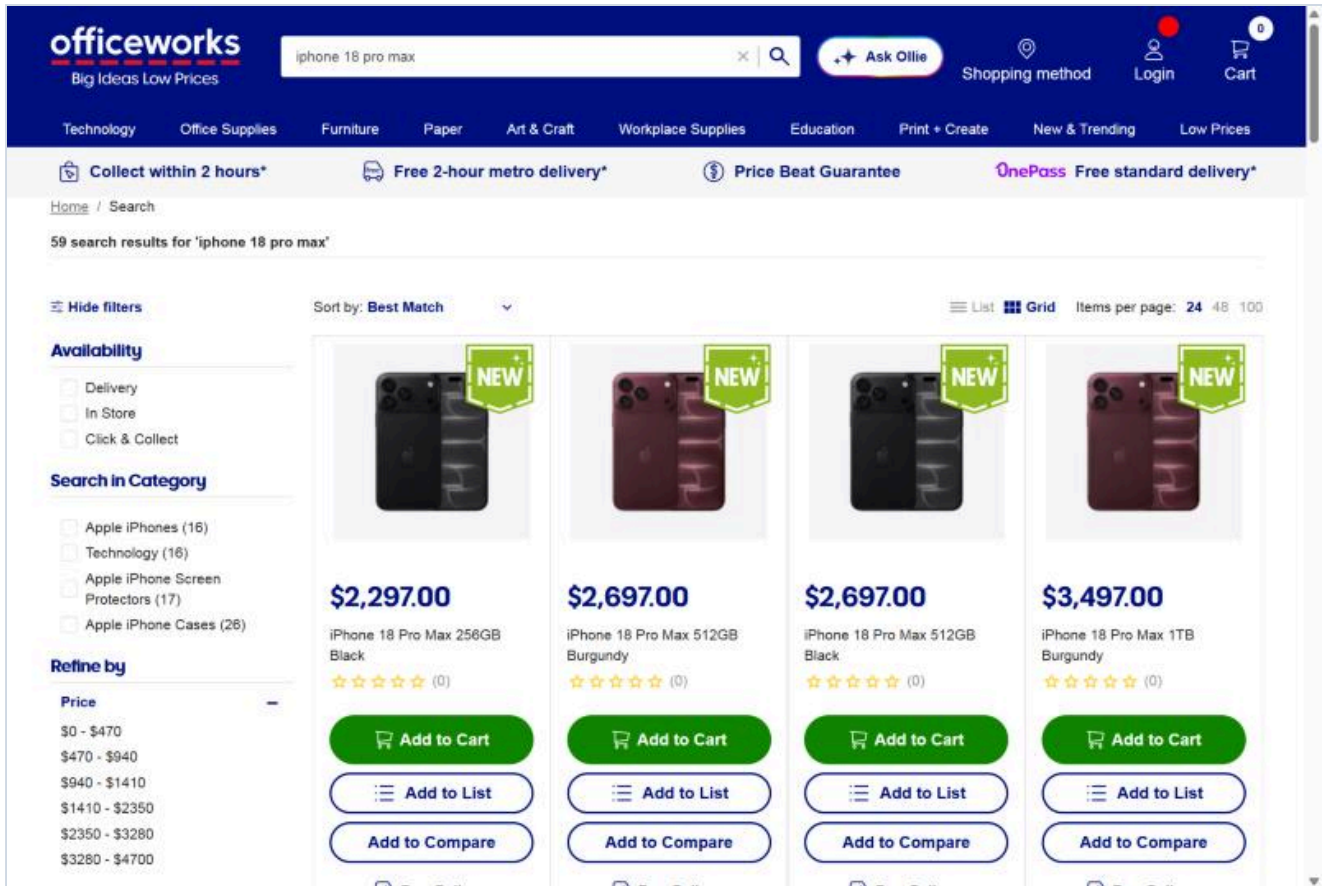


Figure 8 – Officeworks search results: filters (Availability, Category, Price), sort, List/Grid, Add to List / Add to Compare

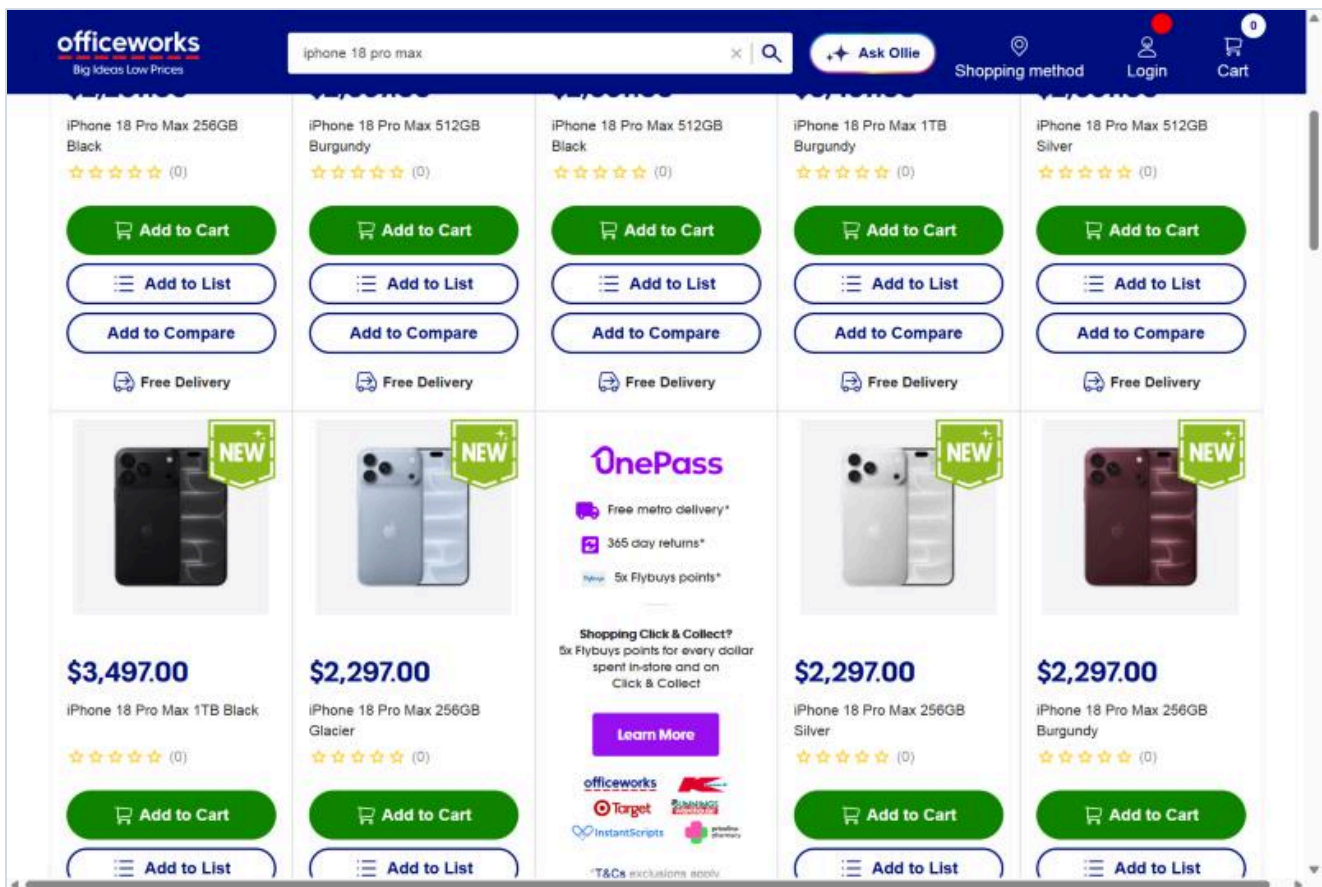


Figure 9 – Officeworks: 'Free Delivery' on product cards and OnePass promotion (free delivery, 365-day returns, Flybuys)

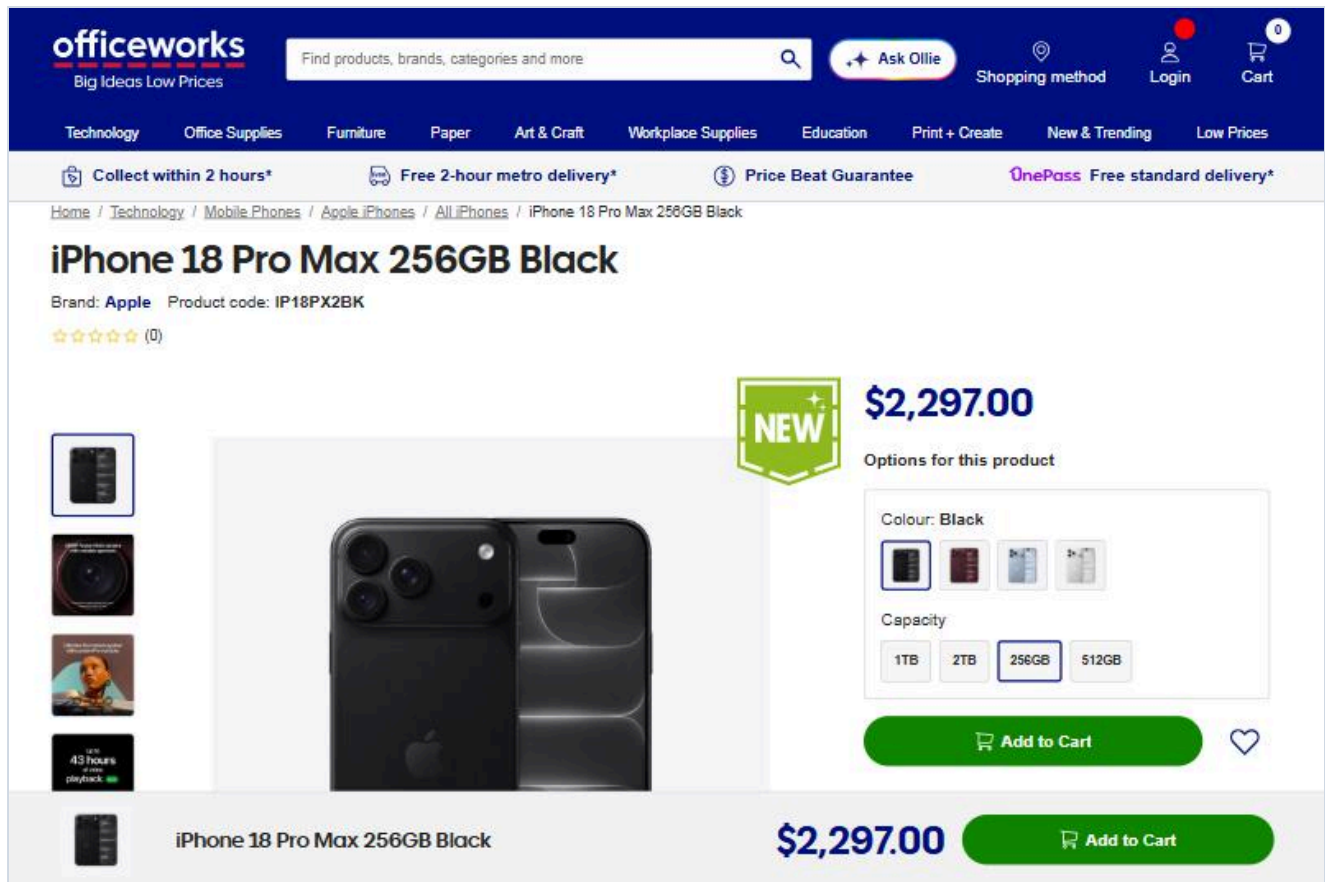


Figure 10 – Officeworks product page: breadcrumbs, product code, price, colour and capacity options, wishlist heart

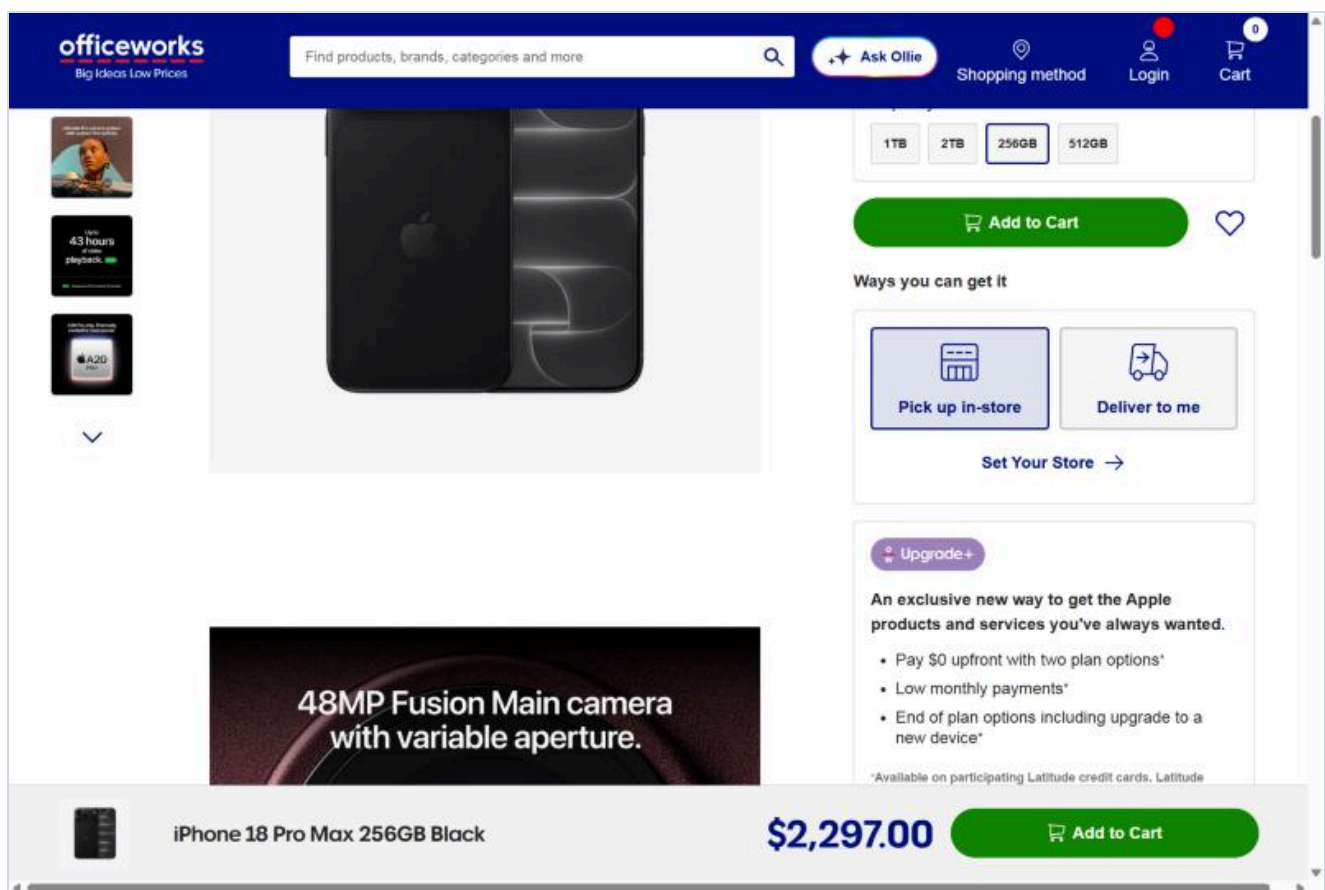


Figure 11 – Officeworks: 'Ways you can get it' (Pick up in-store / Deliver to me) and Upgrade+ payment plan

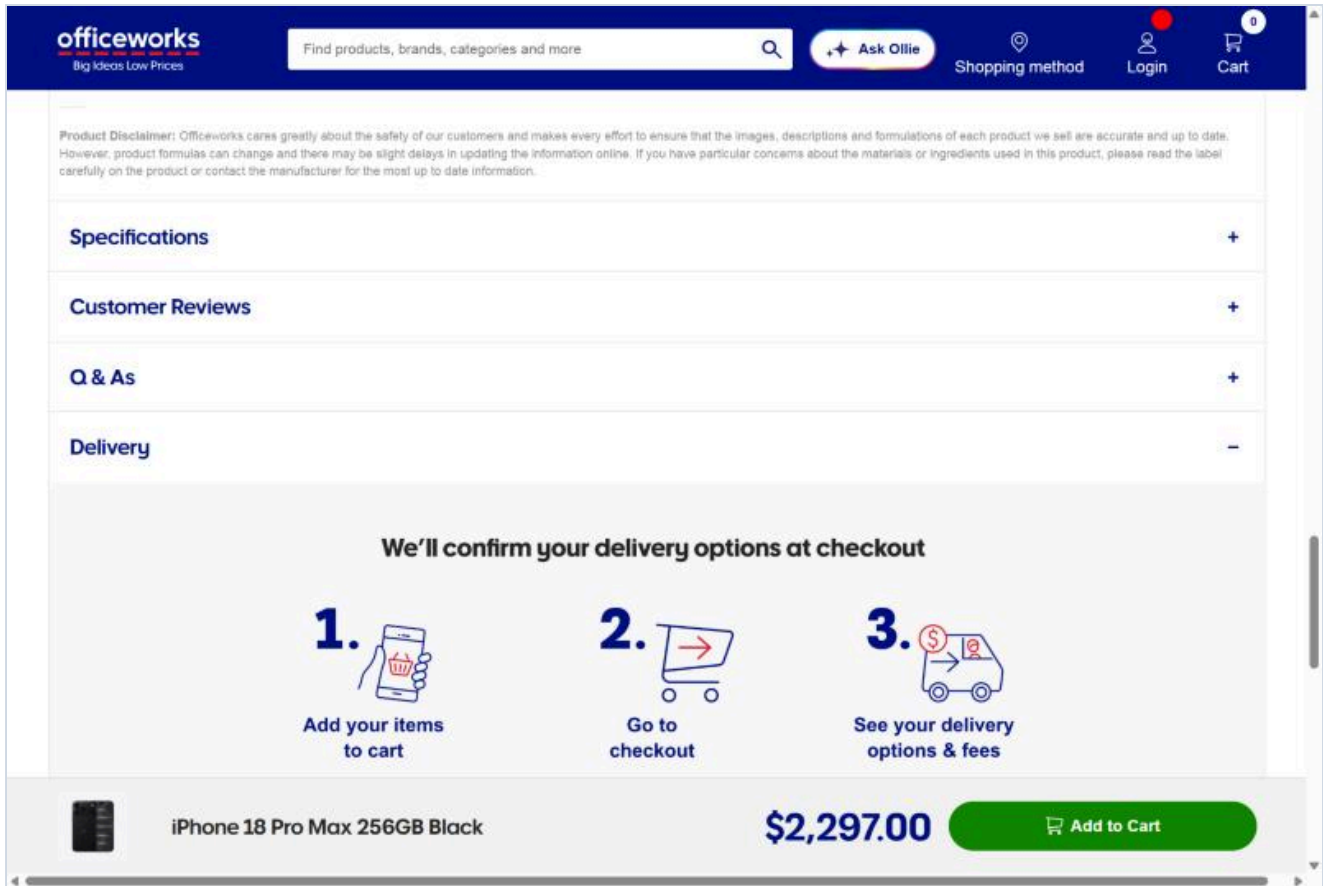


Figure 12 – Officeworks: Specifications, Customer Reviews, Q & As and Delivery sections

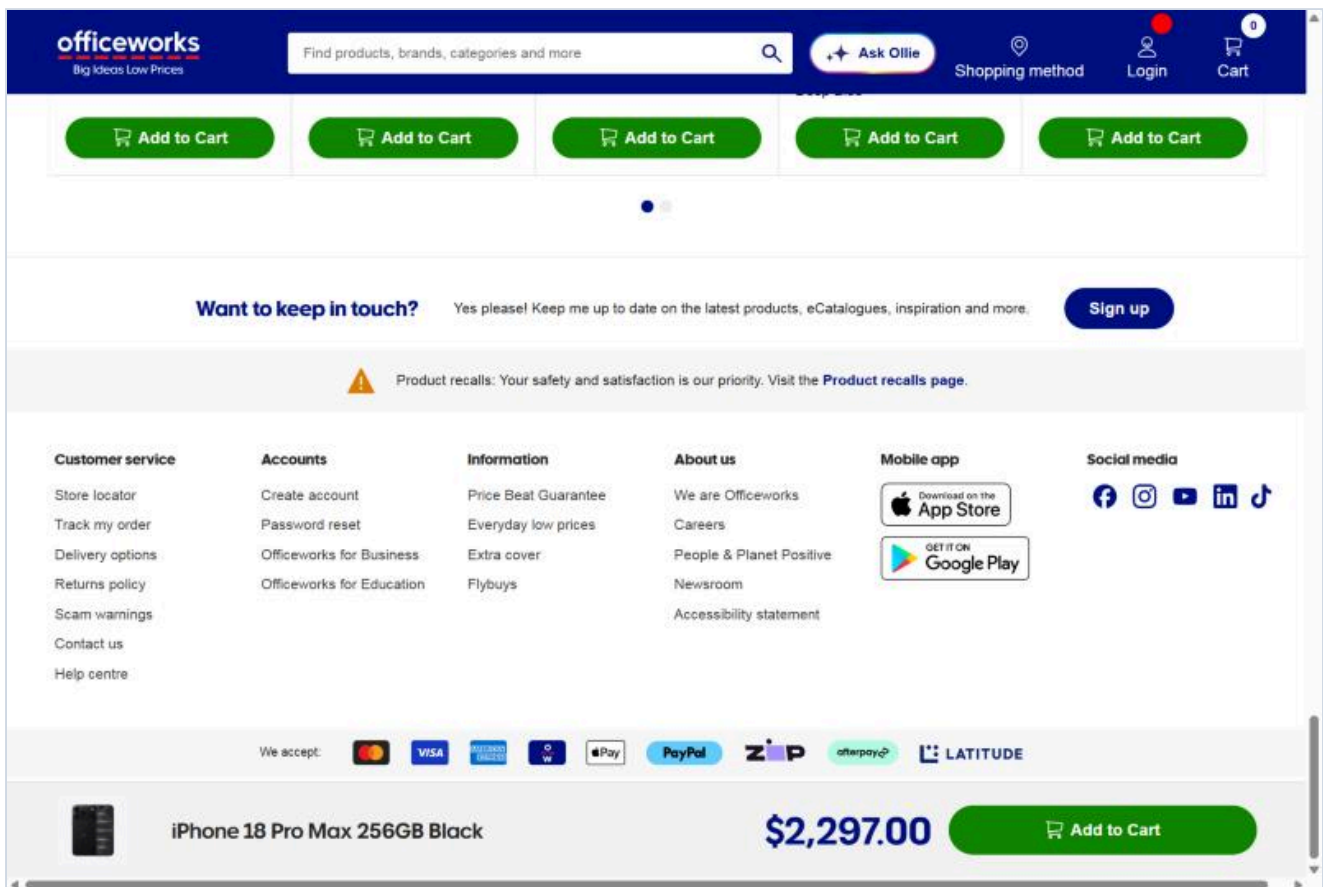


Figure 13 – Officeworks footer: Track my order, delivery options, returns policy, account links, Price Beat Guarantee, payments

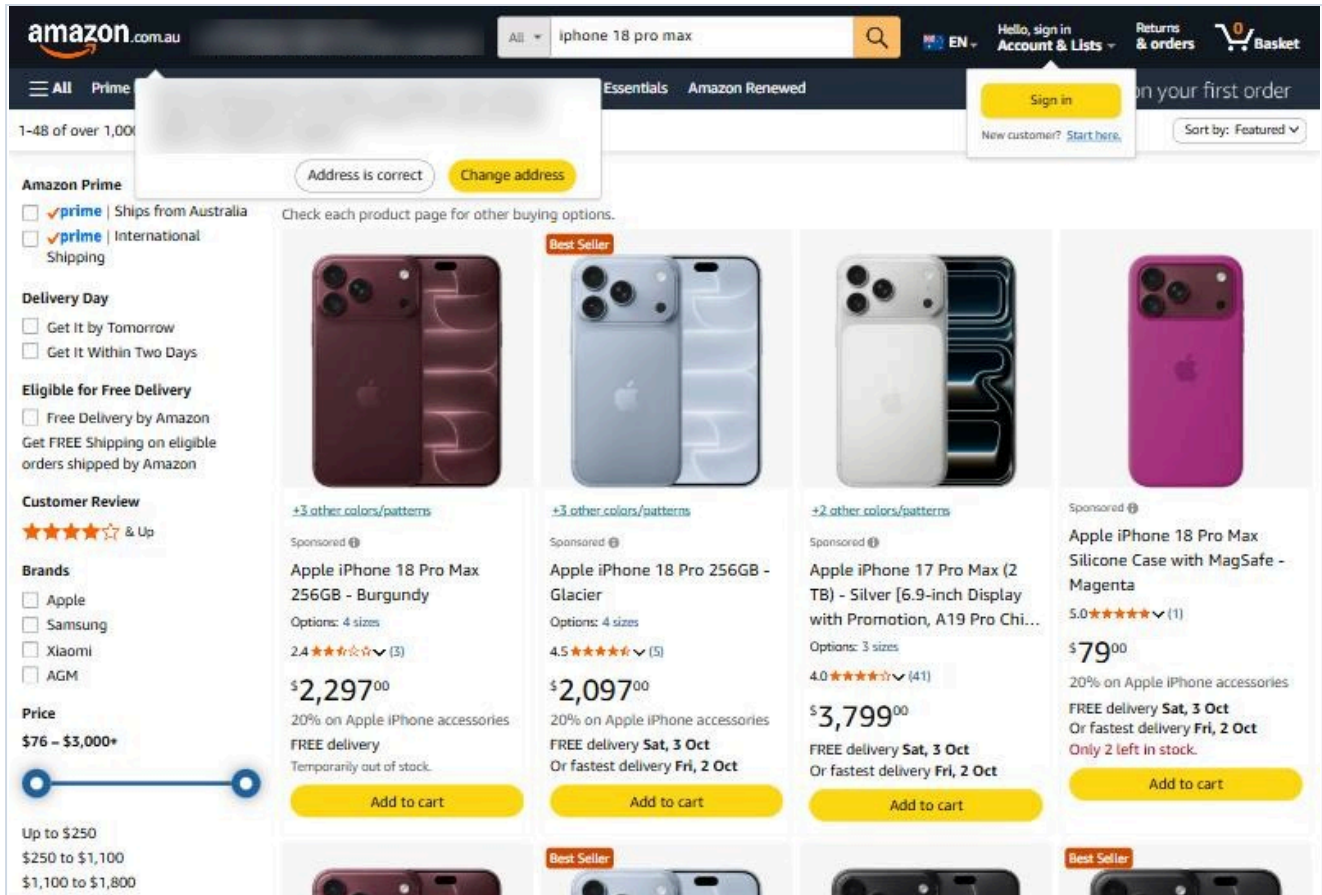


Figure 14 – Amazon search results: filters, sponsored results, Best Seller badges, delivery dates, stock messages (location blurred)

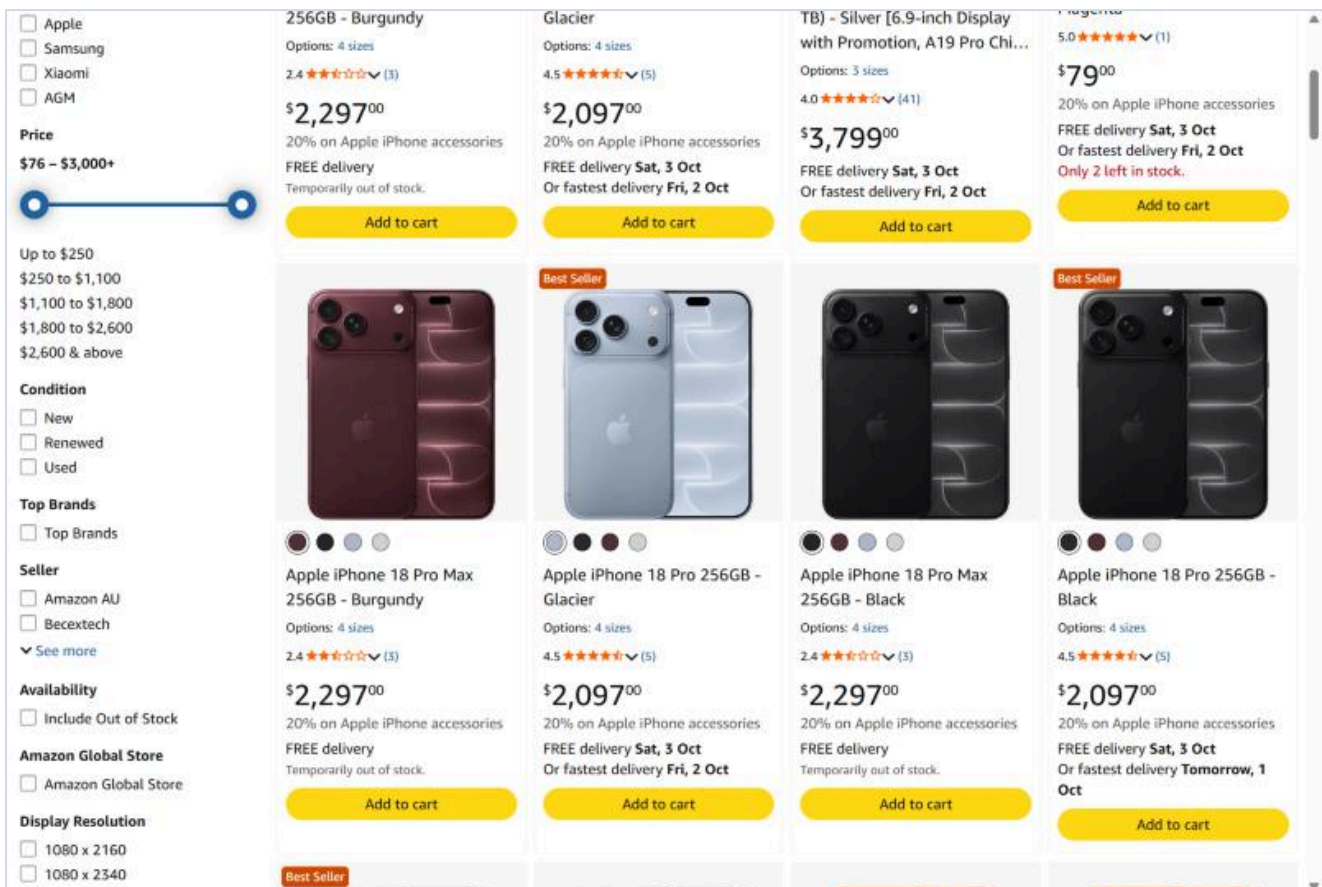


Figure 15 – Amazon: more filters (Condition, Seller, Availability, Display Resolution) and colour swatches

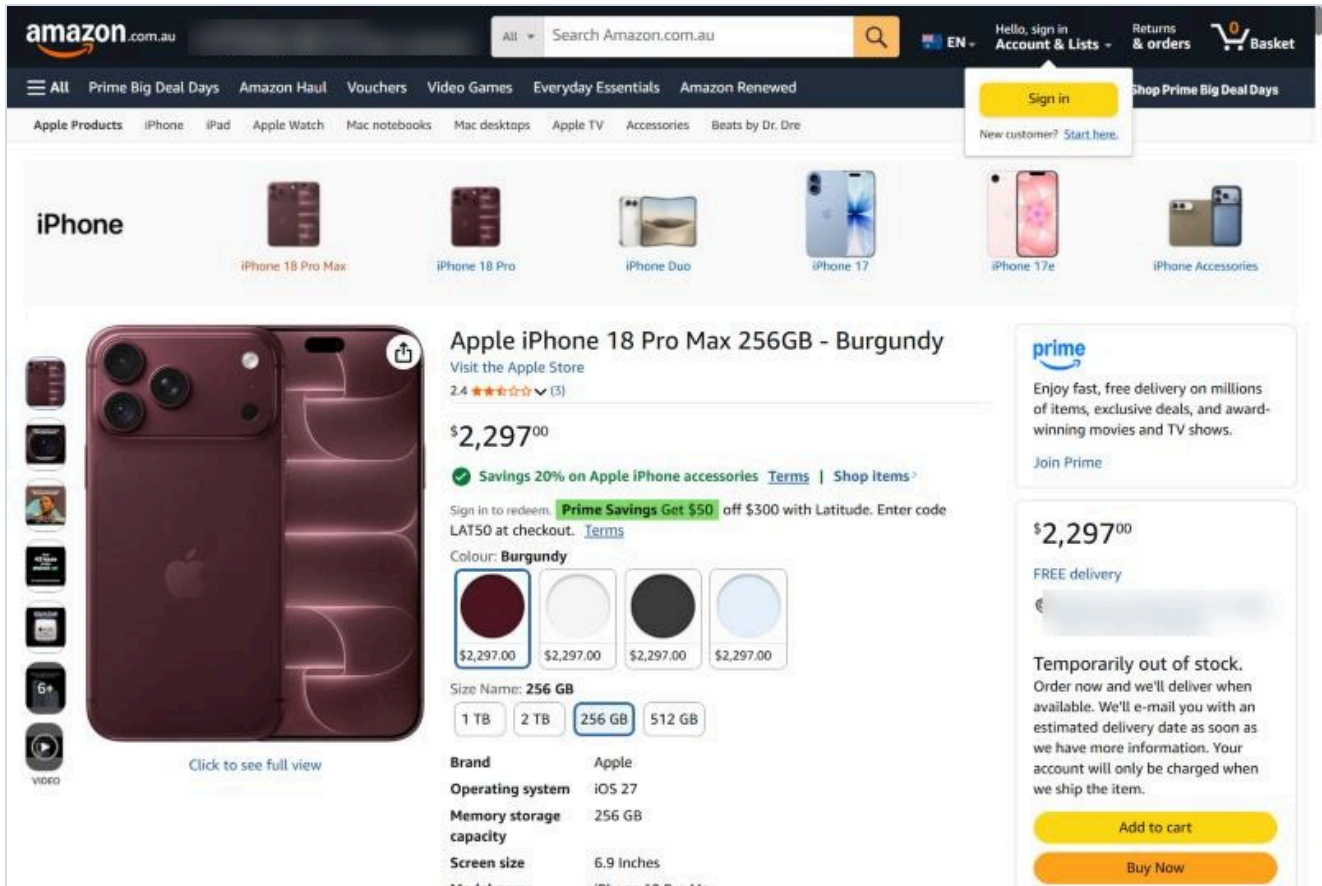


Figure 16 – Amazon product page: rating, price, promotions, colour and size options, key specs, buy box (location blurred)

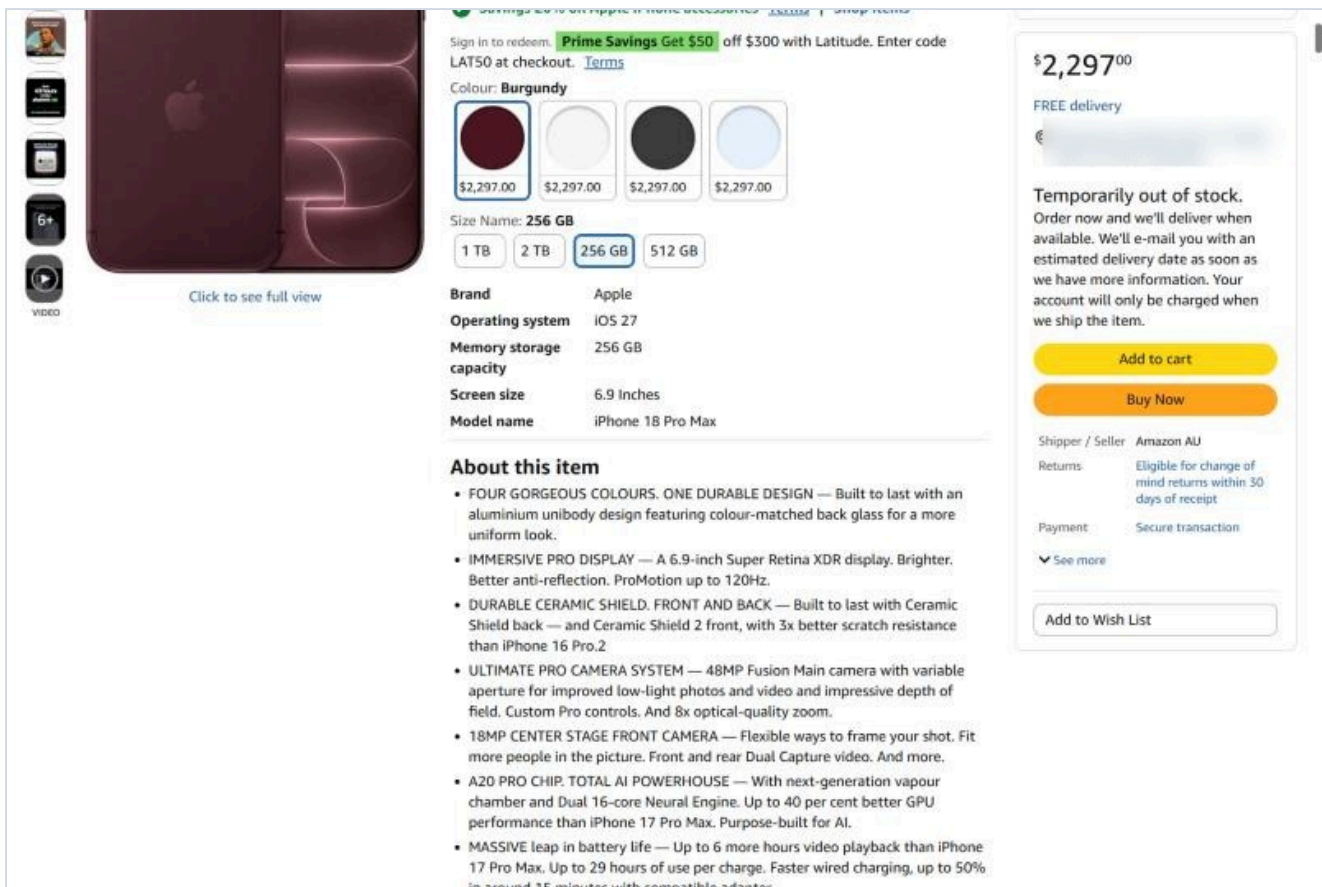


Figure 17 – Amazon buy box: out-of-stock pre-order, Add to cart / Buy Now, seller, 30-day returns, secure transaction, Add to Wish List

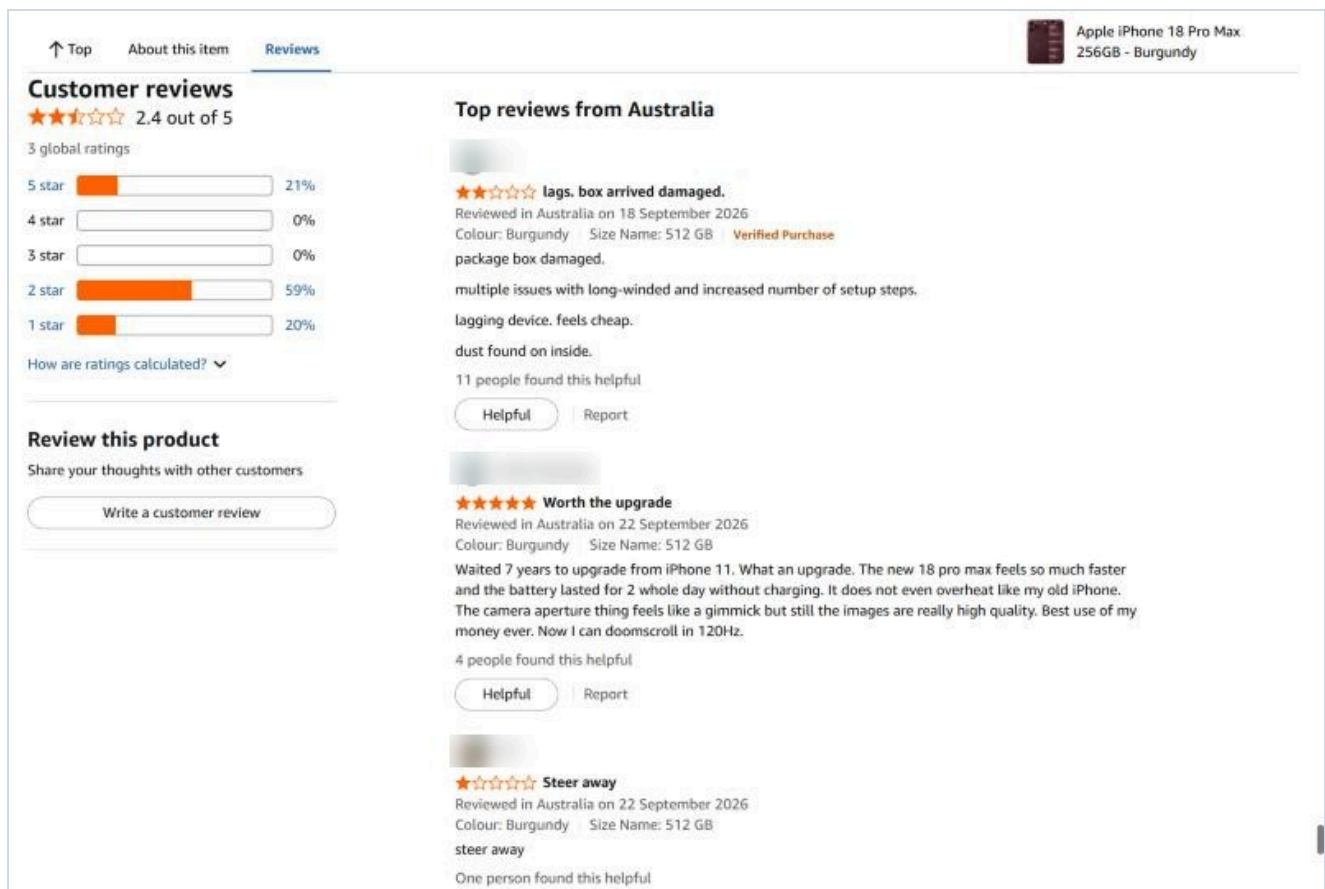


Figure 18 – Amazon customer reviews: 2.4 average with star breakdown, Verified Purchase, Helpful and Report (reviewer names blurred)